Homework 1

Ema Ferrão 109247
Pedro Aldeia 109989

(1)

$\cdot\ H(z) = \sum_{v \in z} P(v) \cdot \log_2\left(\frac{1}{P(v)}\right)$

$\cdot\ H(z|y) = \sum_{v \in y} P(v) \cdot H(z \mid y = v)$

$\cdot\ IG(z|y) = H(z) - H(z|y)$

$$H(y_{out} \mid y_1 > 0,4) = \sum_{v \in y_{out}} P(v) \cdot \log_2\left(\frac{1}{P(v)}\right) =$$

$$= \frac{1}{4} \times \log_2(4) + \frac{3}{8} \times \log_2\left(\frac{8}{3}\right) + \frac{3}{8} \log_2\left(\frac{8}{3}\right) \approx 1,5613$$

$$IG(y_{out} \mid y_1 > 0,4 \wedge y_2) = H(y_{out} \mid y_1 > 0,4) - \sum_{v \in y_2} P(v) \cdot H(y_{out} \mid y_1 > 0,4 \wedge y_2 = v)$$

$$= 1,5613 - \frac{1}{8} \times 0 - \frac{1}{2} \times \left(\frac{1}{2} + \frac{1}{2} + \frac{1}{2}\right) - \frac{3}{8} \times \left(\frac{1}{3} \times \log_2(3) + \frac{2}{3} \log_2\left(\frac{3}{2}\right)\right) =$$

$$\approx 1,5613 - 1,0944 = 0,4669$$

$$IG(y_{out} \mid y_1 > 0,4 \wedge y_3) = 1,5613 - \frac{3}{8} \times \log_2(3) - \frac{1}{4} - \frac{3}{8} \times \log_2(3) =$$

$$= 1,5613 - 1,4387 = 0,1226$$

$$IG(y_{out} \mid y_1 > 0,4 \wedge y_4) = 1,5613 - \frac{1}{8} \times 0 - \frac{3}{8} \times \left(\frac{2}{3} \times \log_2\left(\frac{3}{2}\right) + \frac{1}{3} \times \log_2(3)\right)$$

$$- \frac{1}{2} \times \left(\frac{1}{4} \times \log_2(4) + \frac{3}{4} \times \log_2\left(\frac{4}{3}\right)\right) =$$

$$\simeq 1,5613 - 0,75 = 0,8113$$

$\Rightarrow y_4$ provides the highest gain, hence will be selected.

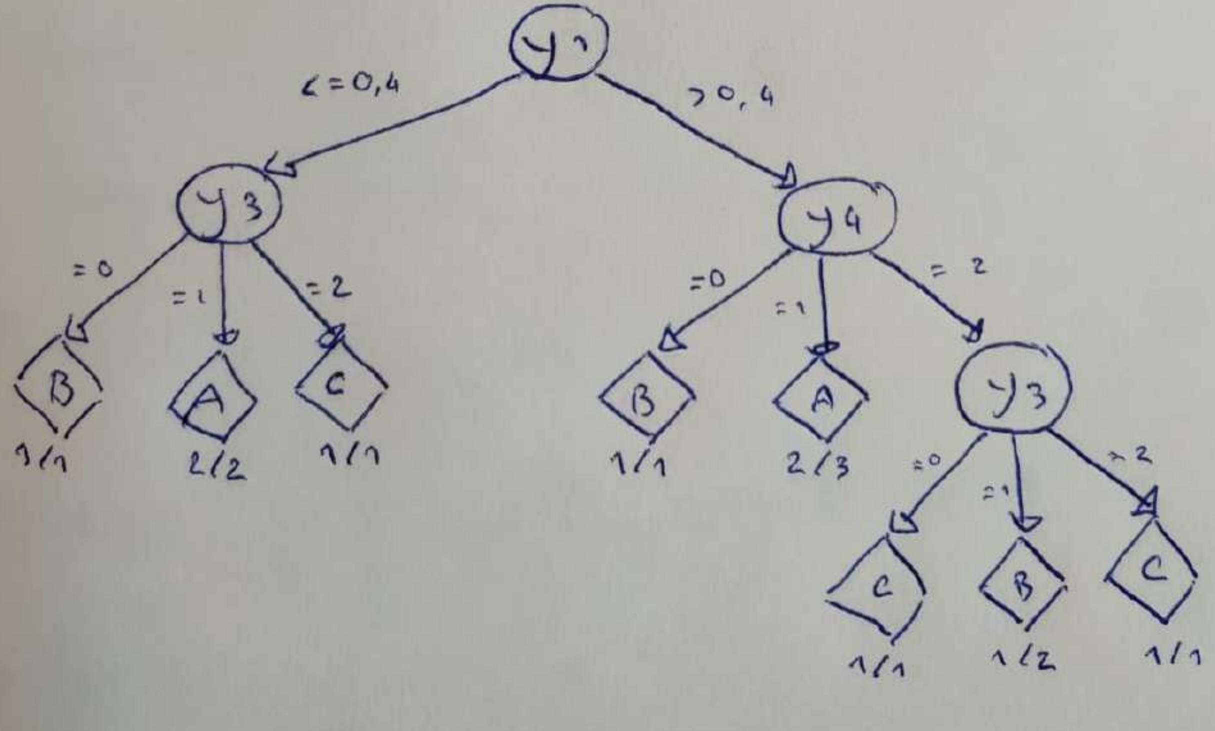
The $y_1 > 0{,}4 \wedge y_4 = 2$ node still has 4 observations so we'll split it.

$$H(y_{out} \mid y_1 > 0{,}4 \wedge y_4 = 2) = \frac{1}{4} \times \log_2(4) + \frac{3}{4} \times \log_2\left(\frac{4}{3}\right) \simeq$$
$$\simeq 0{,}8113$$

$$IG(y_{out} \mid y_1 > 0{,}4 \wedge y_4 = 2 \wedge y_2) =$$
$$= 0{,}8113 - \frac{1}{4} \times 0 - \frac{3}{4} \times \left(\frac{1}{3} \times \log_2(3) + \frac{2}{3} \times \log_2\left(\frac{3}{2}\right)\right) =$$
$$\simeq 0{,}8113 - 0{,}6887 = 0{,}1226$$

$$IG(y_{out} \mid y_1 > 0{,}4 \wedge y_4 = 2 \wedge y_3) = 0{,}8113 - 0 - \frac{1}{2} - 0 =$$
$$= 0{,}3113$$

$\Rightarrow y_3$ provides the highest gain hence will be selected.

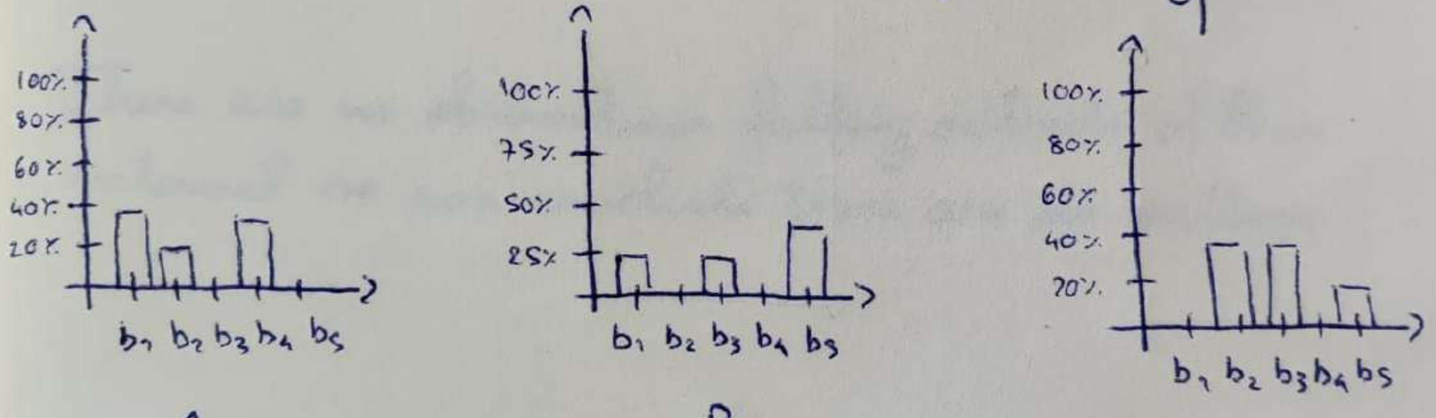


(4)

Observations per class and per bin

	b_1 [0;0,2[	b_2 [0,2;0,4[	b_3 [0,4;0,6[	b_4 [0,6;0,8[	b_5 [0,8;1]	
A	2	1	0	2	0	5
B	1	0	1	0	2	4
C	0	2	2	0	1	5
	3	3	3	2	3	

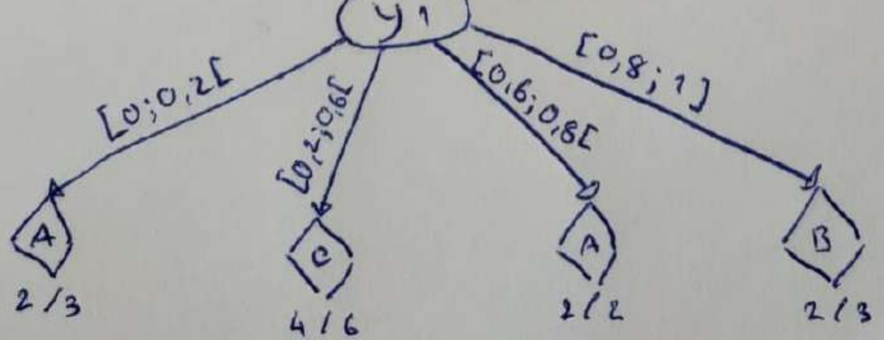
Class conditional relative histograms ~~using~~ of y_1



A B C

Looking at the distribution of classes per bin, we conclude the following classification, which defines the n-ary root split on y_1.

b_1 : A b_2 : C b_3 : C b_4 : A b_5 : B



(2)

Actual

Predicted	A	B	C	
A	4	1	0	5
B	0	3	1	4
C	0	0	3	3
	4	4	4.	

(3)

$$F_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$\text{Precision} = \frac{TP}{TP + FP} \qquad \text{Recall} = \frac{TP}{TP + FN}$$

$$A \longrightarrow P = \frac{4}{5} \quad , \quad R = \frac{4}{4} = 1$$

$$B \longrightarrow P = \frac{3}{4} \quad , \quad R = \frac{3}{4}$$

$$C \longrightarrow P = \frac{3}{3} = 1 \quad , \quad R = \frac{3}{4}$$

$$F_1(A) = 2 \times \frac{\frac{4}{5} \times 1}{\frac{4}{5} + 1} = \frac{8}{9}$$

$$F_1(B) = 2 \times \frac{\frac{3}{4} \times \frac{3}{4}}{\frac{3}{4} + \frac{3}{4}} = \frac{3}{4} \longrightarrow$$

R: B is the class with the lowest F_1 score.

$$F_1(C) = 2 \times \frac{1 \times \frac{3}{4}}{1 + \frac{3}{4}} = \frac{6}{7}$$

(5)

$$Q_1 = \frac{0,25 + 0,33}{2} = 0,29$$

$$Q_3 = \frac{0,71 + 0,83}{2} = 0,77$$

$$IQR = Q_3 - Q_1 = 0,77 - 0,29 = 0,48$$

$$[Q_1 - 1,5 \times IQR ; Q_3 + 1,5 \times IQR] =$$

$$= [0,29 - 1,5 \times 0,48 ; 0,77 + 1,5 \times 0,48] = [-0,43 ; 1,49]$$

There are no observations falling outside of this interval so we conclude there are no outliers.